CS1010 Discrete Mathematics for Computer Science

Counting

Shibam Ghosh

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Department of CSE, IIT Hyderabad



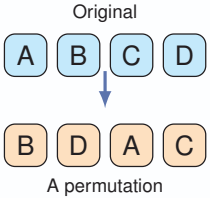
Permutations

What is a Permutation?

Definition

A **permutation** of a set of distinct objects is an **ordered arrangement** of *all* the objects, where each object appears **exactly once**. An ordered arrangement of r elements of a set is called an **r-permutation**.

- Let $A = \{A, B, C, D\}$.
- One permutation is (B, D, A, C).



Observations

- The same objects are used.
- Every object appears exactly once.
- Only the **order** changes.

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A permutation of a set A is a bijection

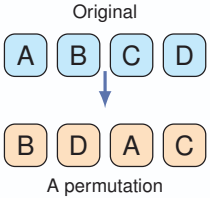
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What is the Number of Possible Permutations?

$\mathcal{Q}$. Consider a set of n distinct objects. What is the number of possible r-permutations where $1 \leq r \leq n$?

$\mathcal{A}$. The first element can be chosen in n ways, the second in $n-1$ ways, and so on until there are $(n-(r-1))$ ways to choose the last element. So, using **the product rule**, the number of possible r-permutations is

$$P(n,r) = n(n-1)(n-2)\dots(n-r+1) = \frac{n!}{(n-r)!}.$$

Note that $P(n,0) = 1$, since there is only one way to order zero elements.

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Practice Questions on Permutations

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Combinations

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- An *r*-**combination** of elements of a set is an unordered selection of *r* elements. Thus, an *r*-combination is simply a subset of size *r*.
- The number of *r*-combinations of a set with *n* distinct elements is denoted by $C(n, r)$ or $\binom{n}{r}$, which is called a **binomial coefficient**.

$\mathcal{Q}$. Show that $C(n, r) = \frac{n!}{(n-r)!r!}$

$\mathcal{A}$. First choose any r objects from the n objects. This can be done in $C(n, r)$ ways. Once these r objects have been chosen, they can be arranged in $P(r, r)$ different orders. Hence, $P(n, r) = C(n, r) \cdot P(r, r)$.

$$\therefore C(n, r) = \frac{P(n, r)}{P(r, r)} = \frac{n!/(n-r)!}{r!/(r-r)!} = \frac{n!}{(n-r)!r!}$$

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$\mathcal{A}$. Since only the set of toppings matters,

$$C(12, 4) = \frac{12!}{4!\,8!} = 495.$$

Corollary

$\mathcal{Q}$. Let n and r be nonnegative integers with $r \leq n$. Then show that $C(n, r) = C(n, n - r)$.

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$\mathcal{A}$. **Proof:** We know from previous results that

$$C(n, r) = \frac{n!}{(n - r)!r!}$$

and

$$C(n, n - r) = \frac{n!}{(n - r)![n - (n - r)]!} = \frac{n!}{(n - r)!r!}$$

Hence, $C(n, r) = C(n, n - r)$.

Q&A: Counting

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$\mathcal{Q}$. In how many ways can a set of two positive integers less than 100 be chosen?

$\mathcal{A}$. There are 99 positive integers less than 100. We choose 2 distinct integers from these 99:

$$\boxed{\binom{99}{2} = 4851}.$$

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$\mathcal{A}$. There are 2^{100} subsets in total. Remove the subsets of size 0, 1, and 2:

$$\boxed{2^{100} - \binom{100}{0} - \binom{100}{1} - \binom{100}{2} = 2^{100} - 5051}.$$

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$\mathcal{A}$. 10 women can be arranged in 10! ways. This creates 11 gaps:

$$_\, W \,_\, W \,_ \cdots W \,_.$$

- Choose 6 of these 11 gaps for the men and arrange the 6 men:

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$\mathcal{Q}$. How many ways are there for four men and five women to stand in a line so that all women stand together?

$\mathcal{A}$. Treat the five women as one block. We now have 5 objects: the women-block and the four men. The 5 objects can be arranged in 5! ways, and the women can be arranged within their block in another 5! ways: $\boxed{5!\,5!}$.

Q&A

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$\mathcal{A}$. Since the committee has 6 members and must have the same number of men and women, we need 3 men and 3 women. Choose 3 men from 10 and 3 women from 15:

$$\boxed{\binom{10}{3}\binom{15}{3} = 120 \cdot 455 = 54\,600}.$$

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$$011, \quad 011, \quad 011, \quad 011, \quad 011.$$



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$\mathcal{A}$. These blocks use 5 zeros and 10 ones. Thus, the remaining number of ones is

$$14 - 10 = 4.$$

We therefore arrange five 011 blocks and four additional 1s. This gives a total of 9 objects, so the number of strings is

$$\boxed{\frac{9!}{5!}} = \boxed{\binom{9}{5}} = \boxed{126}.$$

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$\mathcal{Q}$. How many 4-permutations of the positive integers not exceeding 100 contain three consecutive integers $k, k+1, k+2$ in the correct order, where these integers can perhaps be separated by other integers?

$\mathcal{A}$. There are 98 possible choices for k: $1, 2, \ldots, 98$.

- For each k, choose the fourth integer in $100 - 3 = 97$ ways.
- The three integers can occupy any three of the four positions, while maintaining their order as $C(4, 3) = 4$.
- This gives $98 \cdot 97 \cdot 4$.
- However, a permutation such as $1, 2, 3, 4$ is counted twice: once for $(1, 2, 3)$ and once for $(2, 3, 4)$. There are 98 such overlapping permutations.

Hence,

$$\boxed{98 \cdot 97 \cdot 4 - 98 = 37\,926}.$$

Binomial Coefficients and Some Identities

Expanding $(x + y)^3$

How to show: $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$?

Term	Choose x from	Coefficient
x^3	3 positions	$\binom{3}{3} = 1$
x^2y	2 positions	$\binom{3}{2} = 3$
xy^2	1 position	$\binom{3}{1} = 3$
y^3	0 positions	$\binom{3}{0} = 1$

Let's see the example of x^2y: Different choices of which factor contributes: $(y, x, x), (x, y, x), (x, x, y)$

The Binomial Theorem

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Let x and y be variables, and n a nonnegative integer. Then:

$$(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \cdots + \binom{n}{n} y^n$$

Proof:

$$(x+y)^n = \underbrace{(x+y)(x+y)\cdots(x+y)}_{n \text{ factors}}.$$

Let's count how many $x^{n-j}y^j$, can be obtained from the n factors. To obtain $x^{n-j}y^j$, we must choose j of the n factors to contribute a y. Rest $n-j$ positions for x. Therefore, the coefficient is $\binom{n}{j}$.

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$\mathcal{A}$. The term containing x^3y^2 is obtained by choosing 2 factors contributing $2y$ and the remaining 3 contributing x.

$$\binom{5}{2}x^3(2y)^2 = \binom{5}{2}\cdot 2^2x^3y^2 = 40x^3y^2, \text{ so the coefficient} = 40$$

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$\mathcal{Q}$. Find the coefficient of a^4b^3 in the expansion of $(3a-b)^7$.

$\mathcal{A}$. The term containing a^4b^3 is obtained by choosing 3 factors contributing $(-b)$ and the remaining 4 contributing $3a$.

$$\binom{7}{3}(3a)^4(-b)^3 = -\binom{7}{3}\cdot 3^4a^4b^3 = -2835a^4b^3, \text{ so the coefficient is } -2835.$$

A Useful Identity: Sum of Binomial Coefficients

$\mathcal{Q}$. Show that $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

$\mathcal{A}$. **Proof (using Binomial Theorem):** Let $x = 1$ and $y = 1$:

$$(1+1)^n = \sum_{j=0}^{n} \binom{n}{j} 1^{n-j} 1^j \implies 2^n = \sum_{j=0}^{n} \binom{n}{j}$$

$\mathcal{A}$. **Proof (combinatorial):** Consider the subsets of a set with n elements. There are $\binom{n}{0}$ subsets with 0 elements, $\binom{n}{1}$ with 1 element, ..., and $\binom{n}{n}$ with n elements. The total number of subsets is $\sum_{j=0}^{n} \binom{n}{j}$. Since a set with n elements has 2^n subsets, the identity holds.

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Pascal's Identity

If n and k are integers with $n \geq k \geq 0$, then

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

Proof: Let T be a set with $|T| = n+1$, let $a \in T$, and let $S = T - \{a\}$. There are $\binom{n+1}{k}$ subsets of T containing k elements. Each subset either:

- Contains a with $k-1$ other elements from S: $\binom{n}{k-1}$ ways.
- Contains k elements of S and does not contain a: $\binom{n}{k}$ ways.

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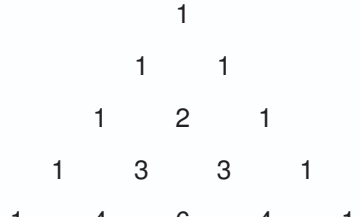
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Pascal's Triangle and the Binomial Theorem

Pascal's Triangle



Each entry is the sum of the two entries directly above it.

Binomial Expansions

$$(x+y)^0 = 1$$

$$(x+y)^1 = x + y$$

$$(x+y)^2 = x^2 + \mathbf{2}xy + y^2$$

$$(x+y)^3 = x^3 + \mathbf{3}x^2y + \mathbf{3}xy^2 + y^3$$

$$(x+y)^4 = x^4 + \mathbf{4}x^3y + \mathbf{6}x^2y^2 + \mathbf{4}xy^3 + y^4$$

Vandermonde's Identity

Let m, n, r be nonnegative integers, with r not exceeding either m or n. Then

$$\sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k} = \binom{m+n}{r}$$

- Suppose there are m items in one set and n items in another set. We want to choose r items from their union.
- **First Way:** The two sets together contain $m+n$ items. Hence, the number of ways to choose r items is $\binom{m+n}{r}$.
- **Second Way:** Another way is to choose k items from the second set. Then we must choose $r-k$ items from the first set. The number of ways to do this is $\binom{n}{k}\binom{m}{r-k}$.
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Another Important Identity

Let n and r be nonnegative integers with $r \leq n$. Then

$$\sum_{j=r}^{n} \binom{j}{r} = \binom{n+1}{r+1}$$

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- Consider the set $\{1, 2, \ldots, n+r+1\}$. We want to choose $r+1$ elements from this set. The total number of choices is $\binom{n+r+1}{r+1}$.
- Classify each $(r+1)$-element subset according to its **largest element**. Suppose the largest element is $r+k+1$, where $k = 0, 1, \ldots, n$.
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$\mathcal{A}$. Combining the two inequalities,

$$\boxed{\binom{n}{k} \leq \frac{n^k}{k!} \leq \frac{n^k}{2^{k-1}}}.$$

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$\mathcal{A}$. By the Binomial Theorem,

$$\left(x + \frac{1}{x}\right)^{100} = \sum_{j=0}^{100} \binom{100}{j} x^{100-j} \left(\frac{1}{x}\right)^j.$$

- Thus, the exponent of x is $100 - j - j = 100 - 2j$. For this to equal k, $j = \frac{100-k}{2}$.
- Hence, the coefficient of x^k is

$$\binom{100}{(100-k)/2}$$

provided k is even and $-100 \leq k \leq 100$; otherwise the coefficient is 0.

Generalized Permutations and Combinations

Permutations with Repetition

Show that the number of r-permutations of a set of n objects with repetition allowed is n^r.

Proof: There are n ways to select an element of the set for each of the r positions in the r-permutation when repetition is allowed. Hence, by the product rule there are n^r ways.

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Combinations with Repetition

2. An ice cream shop offers **5 flavors**:

Vanilla, Chocolate, Strawberry, Mango, Mint.

You would like to buy an ice cream with **8 scoops**. A flavor may be chosen multiple times. How many different ice creams are possible?

Example (Put the scoops into the corresponding flavor box.):

3	2	0	1	2
V	C	S	M	Mt

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Combinations with Repetition (Cont.)

$$\boxed{\text{Vanilla} = 3,\ \text{Chocolate} = 2,\ \text{Strawberry} = 0,\ \text{Mango} = 1,\ \text{Mint} = 2},$$

Represent each scoop by a $\star$ and separate flavors using bars.

$$\star\star\star|\ \star\star|\ |\ \star\ |\ \star\star$$

$\mathcal{A}$. There are 8 stars and 4 bars. Hence, every ice cream corresponds to an arrangement of $8 + 4 = 12$ symbols. Therefore,

$$\binom{12}{4} = \binom{12}{8} = 495.$$

Combinations with Repetition

General Rule

The number of ways to choose n objects from k types, where repetition is allowed:

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}.$$

- Suppose the numbers of selected objects of each type are $x_1, x_2, \ldots, x_k$:

$$x_1 + x_2 + \cdots + x_k = n, \qquad x_i \geq 0.$$

- Each solution can be represented by: $\underbrace{\star \cdots \star}_{x_1} \mid \underbrace{\star \cdots \star}_{x_2} \mid \cdots \mid \underbrace{\star \cdots \star}_{x_k}$.
- Thus, there are n stars and $k-1$ bars and every selection corresponds to an arrangement of $n+k-1$ symbols.
- Choosing the positions of the n stars (or equivalently the $k-1$ bars) gives

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}.$$



Thank You for your attention!

Any questions?